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When personalization deters delegation to algorithms: Experimental evidence

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- ▶ Huge & unconnected literature on algorithm/automation/AI/... reliance
- ▶ Most of the literature: setups with clearly identifiable mistakes
- ▶ But in many real-life settings, preferences determine what is a mistake
- ▶ E.g., robo-advisors: maximal payoff vs best fit to risk-parameters



- ▶ Does algorithm reliance differ for **objective optimization vs. preference-based personalisation** in a preferential-choice environment?
 - (i) ...at first contact?
 - (ii) ...overall?



- ▶ What do we mean by **objective optimization**?
 - an algorithm that chooses some ‘objectively ideal’ option independently of a subject’s preferences (here: **highest EV**)
- ▶ What do we mean by **personalised**?
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 - an algorithm that measures a subject’s preferences and chooses accordingly (here: **implementation of revealed preferences**)
- ▶ In the experiment, we describe both algorithms in intuitive terms:

EV: *‘The algorithm works out how much each option would yield on average and selects the option for which this is highest.’*

Prefs: *‘The algorithm analyses your choices from Part 1 and selects the best option for you in light of your Part-1 choices.’*

Well-known (Swiss) robo-advisors(' marketing strategies)

- ▶ “We rely on a low-cost, passive investment strategy and have selected a total of 9 equity, bond and real estate ETFs from over 1,500 ETFs (exchange-traded funds) and put them together in such a way that they **achieve an optimal risk/return ratio.**” (findependent.ch; for GER, see e.g., whitebox.eu)
- ▶ “Chat with Selma: Selma **looks at your financial life and learns how you should structure your finances.** Get your plan: Selma **creates a personalized investment plan that fits your life.** Sit back and relax! Selma takes care of your money from now on and does all the buying and selling for you.” (selma.com; for GER, see e.g., quirion.de)

Part 1 – Elicitation of preferences

- only two, ‘well-presented’ lotteries, unlimited time
- “hypothetical, but the choices may be meaningful for Part 2”

Part 2 – Lottery Selection Task

- could stand for an investment task
- complexity (number of options, outcomes under different SOTW)
- time pressure (think: relative to the number of options)
- each Part-2 task built from 2 anchor options (→ Part 1) + 6 decoys

	State 1	State 2	State 3	State 4	State 5	State 6	State 7	State 8	State 9	State 10
Option 1	225	225	225	225	225	225	225	205	0	0
Option 2	525	525	525	525	525	0	0	0	0	0

State:	1	2	3	4	5	6	7	8	9	10
Option 1	225	225	0	225	0	225	225	225	225	0
Option 2	525	0	505	0	525	0	0	0	525	505
Option 3	525	0	525	0	525	0	0	0	525	525
Option 4	225	225	0	225	205	205	205	225	225	0
Option 5	225	205	0	205	205	225	225	205	225	0
Option 6	225	225	0	225	205	225	225	225	225	0
Option 7	525	0	525	0	525	0	0	0	225	525
Option 8	505	0	525	0	505	0	0	0	505	525

- For further distraction: 4 filler tasks in between
- 8 rounds: 5' pre-view — 5' algorithm choice—8' decision/waiting time

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- ▶ **Recruitment:** Prolific UK, online (z-Tree); preregistered target 120/treatment before exclusions.
- ▶ **Exclusions (preregistered):** bot/slider checks, comprehension after 2nd attempt, self-reported AI use, super-fast log-times, incomplete data.
- ▶ **Final sample:** **N = 233** (EV: $n = 118$, Prefs: $n = 115$).
- ▶ **Incentives:** GBP 2.00 base + one randomly drawn Part-2 round (1 penny / point); up to GBP 8.90 bonus. Mean earning GBP 4.14, median duration ~16 min.



- ▶ **H1.** People are more likely to delegate financial decisions to an **EV-maximising** algorithm than to a **personalised** algorithm that incorporates their (revealed) preferences.
- ▶ **Primary tests (preregistered):**
 - (1) One-sided **Boschloo** exact test on **Round-1** delegation (first contact)
 - (2) One-sided **WMW** test on participant-level mean delegation across Part 2
- ▶ **Preregistered LPM:** decision-set FE, participant-clustered SE, treatment $\times$ period interaction, share of prior delegated rounds with zero payoff.
- ▶ **Exploratory:** categorical *never-delegation*, win-stay/lose-shift, ex-post reason items.



- ▶ **Benchmark:** only ~34–35% of *self-chosen* options were non-dominated; delegated rounds paid on average ~25% **more** than self-decided ones $\Rightarrow$ delegation is the instrumentally rational thing to do.
- ▶ Round 1: 66.9% (EV) vs. 55.8% (Prefs).
- ▶ Boschloo exact test, one-sided: $p = 0.0413 \Rightarrow$ **H1 supported at first contact.**
- ▶ Treatment gap concentrated in the first decision.

Main Results: First Contact



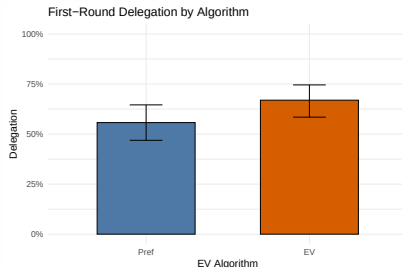
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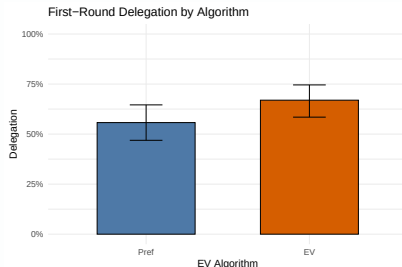


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Main Results: Repeated Decisions

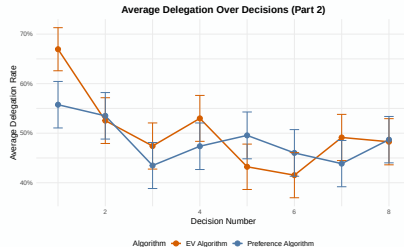


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- ▶ **Average delegation:** no significant treatment difference overall (WMW one-sided $p = 0.2425$).
- ▶ Preregistered LPM (decision-set FE, clustered SE):
 - EV main effect: $\beta = -0.049$, $p = 0.56$
 - EV $\times$ period: $\beta = 0.002$, $p = 0.87$
 - Share of prior zero-payoff delegated rounds:
 $\beta = -0.385$, $p < 0.001$
- ▶ Convergence after Round 1 driven **more by a decline in EV** than by a rise in Prefs.
- ▶ But: first-contact decisions can determine whether users *come back at all*.



Categorical Non-Delegation (not preregistered)

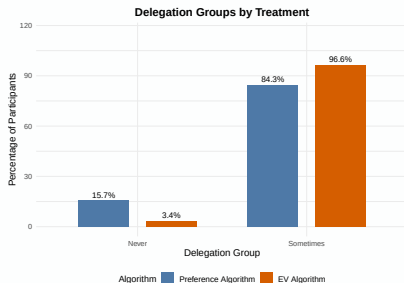


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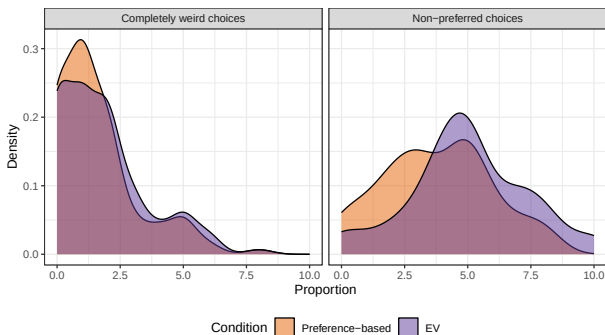
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- ▶ **Never-delegators:**
 - **15.7%** in Prefs (18/115)
 - **3.4%** in EV (4/118)
- ▶ Always-delegators show the opposite, weaker pattern (13.6% Prefs vs. 19.1% EV).
- ▶ Never-delegators do *not* differ in age, income, education, gender, time spent, financial risk-taking, AI usage, or reliability.
- ▶ Within Prefs: never-delegators report (weakly) *higher* satisfaction with Part-1 choices ($\Rightarrow$ not driven by dislike of the elicited profile).
- ▶ **Replication** in earlier (PhD-thesis) study: 32/189 vs. 17/187, Boschloo $p = 0.0135$.



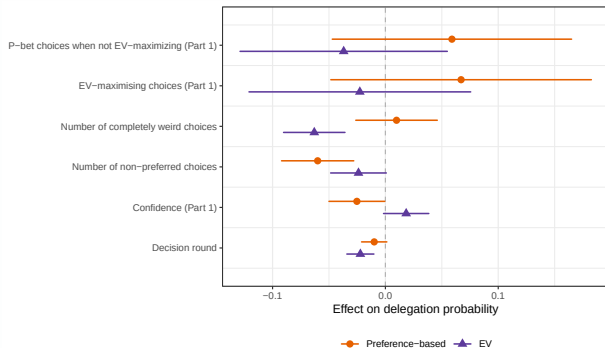
Mechanisms: *What* deters delegation differs by algorithm

- LPM of individual delegation on *expected non-preferred* and *expected “completely weird”* choices, $\times$ treatment, controlling for round, Part-1 confidence, risk preferences, task FE:



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- ▶ LPM of individual delegation on *expected non-preferred* and *expected “completely weird”* choices, $\times$ treatment, controlling for round, Part-1 confidence, risk preferences, task FE:



95% confidence intervals. Linear probability model with task fixed effects and part EV-condition effects are main effect + interaction term.

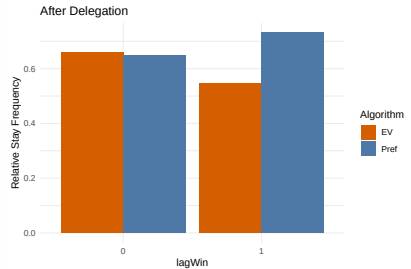
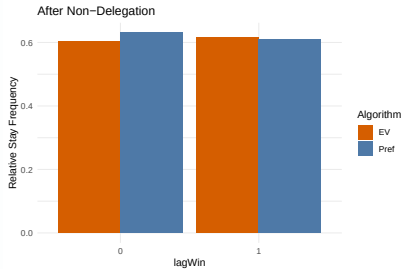
- ▶ Differences run more by *delegator type* than by treatment.
- ▶ **Delegators** (vs. never-delegators) more often cite performance motives (70.1% vs. 36.4%: “computer would choose better”).
- ▶ **Never-delegators** more often cite distrust of the algorithm and responsibility for the choice (63.6% vs. 38.9%).
- ▶ Crucially: they want responsibility for the **choice**, *not* for the **outcome** ⇒ this is a positive valuation of **choice authority**, not blame-avoidance.
- ▶ Items on being algorithmically analysed and on opacity (“don’t know how it chooses”) are rarely selected ⇒ not a transparency story.
- ▶ Consistent with *control* / *decision-autonomy* as a driver of acceptance (Schaap et al., 2024; Ivanova-Stenzel & Tolksdorf, 2025).

Dynamics: win-stay/lose-shift



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1. **Preregistered behavioural** comparison of *objective* vs *personalised* decision rules $\Rightarrow$ personalisation does *not* increase delegation.
2. The effect is concentrated at **first contact** (Round 1: 66.9% EV vs 55.8% Prefs) — and first contact often decides whether users return at all.
3. **Categorical non-delegation** is a distinct uptake barrier under personalisation (15.7% vs 3.4%), *not* explained by demographics, risk attitudes, AI exposure, or dissatisfaction with the elicited profile.

Thank you!

Thank you — and looking forward to your
questions.

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Preregistration: <https://aspredicted.org/7un5qf.pdf>